Def For $x \in \mathbb{R}$, the absolute value of x is

$$|x| = \begin{cases} x & x \geq 0, \\ -x & x < 0. \end{cases}$$

Exc This function satisfies the following properties:

1. $|x| \geq 0 \quad \forall x \in \mathbb{R}$
2. $|x| = 0 \iff x = 0$
3. $|x+y| \leq |x| + |y| \quad \forall x, y \in \mathbb{R}$ (Triangle inequality)
4. $\big||x| - |y|\big| \leq |x-y| \quad \forall x, y \in \mathbb{R}$
5. $|x \cdot y| = |x| \cdot |y| \quad \forall x, y \in \mathbb{R}$

Def We say that the sequence $\{a_n\}_{n \geq 1}$ <u>converges</u> to $a \in \mathbb{R}$ if: $\forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < \varepsilon \quad \forall n \geq N.$$

We say that a is the <u>limit</u> of $\{a_n\}_{n \geq 1}$, and we write "$\lim_{n \to \infty} a_n = a$" or "$a_n \to a$ as $n \to \infty$".

We say $\{a_n\}_{n \geq 1}$ <u>diverges</u> if it does not converge.

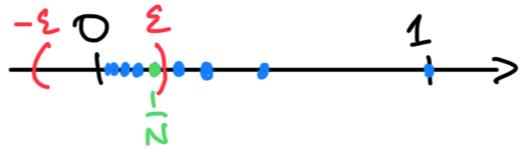
Rmk Note that N depends on ε, but not on n.

Ex Prove that $\lim_{n \to \infty} \frac{1}{n} = 0$.

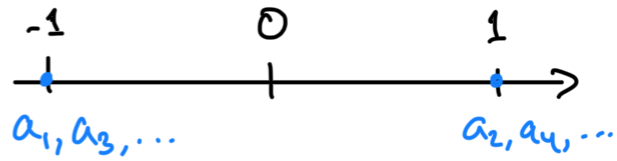
Pf: Fix $\varepsilon > 0$. Then $\exists N \in \mathbb{N}$ s.t. $N > \frac{1}{\varepsilon}$ (by the Archimedean property). So for $n \geq N$ we have

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore $\lim_{n \to \infty} \frac{1}{n} = 0$. $\square$



<u>Ex</u> Prove that the sequence $\{(-1)^n\}_{n\geq 1}$ diverges.



<u>Pf</u>: Suppose not: $\exists\, a \in \mathbb{R}$ s.t. $\lim_{n\to\infty} (-1)^n = a$. Taking $\varepsilon = \frac{1}{2}$ in the definition of limit, we get: $\exists\, N \in \mathbb{N}$ s.t.

$$|(-1)^n - a| < \tfrac{1}{2} \quad \forall\, n \geq N.$$

Taking $n = 2N$, we get $|1-a| < \frac{1}{2}$

$n = 2N+1$, we get $|-1-a| = |1+a| < \frac{1}{2}$.

Together,

$$2 = |1+1| = |1-a+1+a| \leq |1-a| + |1+a|$$
$$< \tfrac{1}{2} + \tfrac{1}{2} = 1,$$

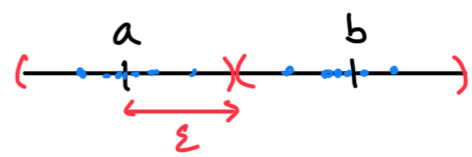
a contradiction. $\square$

<u>Lem</u> The limit of a sequence $\{a_n\}_{n\geq 1}$ is unique, if it exists.

<u>Pf</u>: Suppose $\exists\, a, b \in \mathbb{R}$ s.t. $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} a_n = b$.

Want: $a = b$.

Suppose not: $a \neq b$.



Set $\varepsilon = \frac{|b-a|}{2}$. Then $\varepsilon > 0$,

and so

$$\lim_{n\to\infty} a_n = a \Rightarrow \exists\, N_1 \in \mathbb{N} \text{ s.t. } |a_n - a| < \varepsilon \ \forall\, n \geq N_1$$

$$\lim_{n\to\infty} a_n = b \Rightarrow \exists N_2 \in \mathbb{N} \text{ s.t. } |a_n - b| < \varepsilon \quad \forall n \geq N_2$$

Set $N = \max\{N_1, N_2\}$. Then for $n \geq N$ we have

$$|b-a| = |b - a_n + a_n - a| \leq |b - a_n| + |a_n - a|$$
$$< \varepsilon + \varepsilon = 2\varepsilon = |b-a|,$$

a contradiction. $\square$

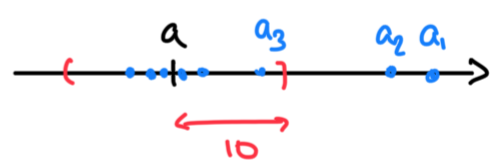
<u>Lem</u> If $\{a_n\}_{n\geq 1}$ converges, then it is bounded.

<u>Def</u> We say a sequence $\{a_n\}_{n\geq 1}$ is <u>bounded above</u> / <u>bounded below</u> / <u>bounded</u> if the set $\{a_n : n \geq 1\}$ is bounded above / bounded below / bounded.

<u>Pf</u>: As $\{a_n\}_{n\geq 1}$ converges, $\exists a \in \mathbb{R}$ s.t. $\lim_{n\to\infty} a_n = a$.
Taking $\varepsilon = 10$ in the definition of limit, we get:
$\exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < 10 \quad \forall n \geq N.$$

So, for $n \geq N$ we have



$$|a_n| \leq |a_n - a| + |a| < 10 + |a|.$$

Set $M = \max\{10 + |a|, |a_1|, |a_2|, \ldots, |a_{N-1}|\}$. Then

$$|a_n| \leq M \quad \forall n \geq 1,$$

so M is an upper bound and $-M$ is a lower bound. Therefore $\{a_n\}_{n\geq 1}$ is bounded. $\square$

<u>Thm</u> If the sequence $\{a_n\}_{n\geq 1}$ converges to $a \in \mathbb{R}$,

then $\forall k \in \mathbb{R}$ the sequence $\{k \cdot a_n\}_{n \geq 1}$ converges and

$$\lim_{n \to \infty} k a_n = k \cdot a.$$

<u>Pf</u>: Fix $k \in \mathbb{R}$. Let $\varepsilon > 0$. As $\lim_{n \to \infty} a_n = a$, $\exists N \in \mathbb{N}$ s.t. $\forall n \geq N$ we have

$$|a_n - a| < \frac{\varepsilon}{|k|+1} \leftarrow \neq 0 \quad \forall k$$

- $\frac{\varepsilon}{|k|}$ does not work, because we could have $k=0$.

Then, for $n \geq N$ we have

$$|ka_n - ka| = |k| \cdot |a_n - a| \leq |k| \cdot \frac{\varepsilon}{|k|+1} < \varepsilon.$$

So $\lim_{n \to \infty} k a_n = ka$. $\square$

<u>Thm</u> If the sequences $\{a_n\}_{n \geq 1}$ and $\{b_n\}_{n \geq 1}$ converge to a and b respectively, then $\{a_n + b_n\}_{n \geq 1}$ converges and

$$\lim_{n \to \infty} (a_n + b_n) = a + b.$$

<u>Pf</u>: Fix $\varepsilon > 0$. As $\lim_{n \to \infty} a_n = a$, $\exists N_1 \in \mathbb{N}$ s.t.

$$|a_n - a| < \frac{\varepsilon}{2} \qquad \forall n \geq N_1.$$

As $\lim_{n \to \infty} b_n = b$, $\exists N_2 \in \mathbb{N}$ s.t.

$$|b_n - b| < \frac{\varepsilon}{2} \qquad \forall n \geq N_2.$$

Set $N = \max\{N_1, N_2\}$. Then for $n \geq N$ we have

$$\begin{aligned} |(a_n + b_n) - (a+b)| &\leq |a - a_n| + |b - b_n| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

So $\lim_{n \to \infty} (a_n + b_n) = a + b.$ $\square$